

Characteristics: Burgers vs Wave

Geometry, Breakdown, and d'Alembert's Formula

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Two Prototype Equations

Burgers' Equation

$$u_t + uu_x = 0$$

Wave Equation

$$u_{tt} - c^2 u_{xx} = 0$$

- Both propagate information
- Both admit characteristic curves
- Geometry is radically different

Order and Structure

Burgers	Wave
First order	Second order
One family of curves	Two families
Nonlinear	Linear
Speed depends on u	Speed fixed $\pm c$

The order of the equation determines the geometry.

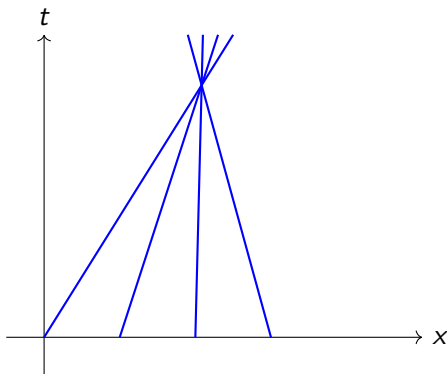
Burgers: Characteristic ODEs

$$\frac{dx}{dt} = u, \quad \frac{du}{dt} = 0$$

$$x(t) = x_0 + u_0(x_0)t$$

- Speed determined by initial data
- Faster characteristics overtake slower ones

Burgers: Crossing Characteristics



Characteristics intersect $\Rightarrow$ shock formation.

Wave Equation Factorization

$$u_{tt} - c^2 u_{xx} = (\partial_t - c\partial_x)(\partial_t + c\partial_x)u$$

Two transport equations emerge.

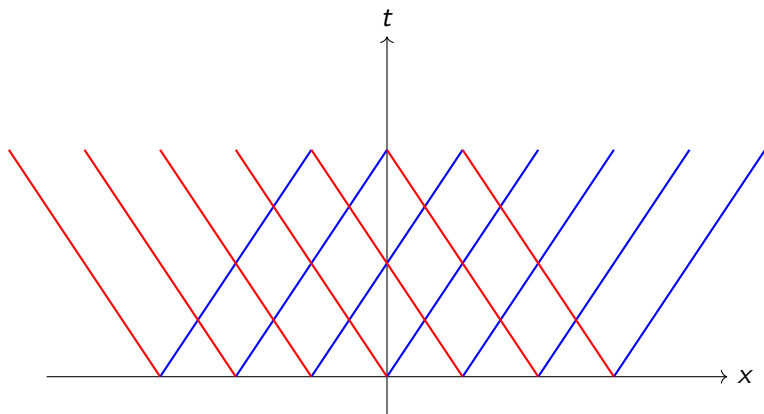
Wave Characteristics

$$\frac{dx}{dt} = \pm c$$

$$x - ct = \text{constant}, \quad x + ct = \text{constant}$$

- Two independent families
- Parallel straight lines
- Never intersect

Wave: Spacetime Picture



Left- and right-moving waves propagate independently.

Fundamental Contrast

Burgers

- One family
- Speed varies
- Curves cross
- Shocks form

Wave

- Two families
- Fixed speed $\pm c$
- No crossing
- Global smooth solutions

Step 1: Characteristic Coordinates

Define new variables:

$$\xi = x - ct, \quad \eta = x + ct$$

These are constant along the two characteristic families.

Step 2: Rewrite Derivatives

Using the chain rule:

$$\partial_x = \partial_\xi + \partial_\eta$$

$$\partial_t = -c\partial_\xi + c\partial_\eta$$

Compute:

$$u_{tt} - c^2 u_{xx} = 4c^2 u_{\xi\eta}$$

Step 3: Reduced Equation

The wave equation becomes

$$u_{\xi\eta} = 0.$$

Integrate with respect to η :

$$u_{\xi} = F(\xi)$$

Integrate again:

$$u(\xi, \eta) = f(\xi) + g(\eta).$$

d'Alembert's Formula

Returning to (x, t) :

$$u(x, t) = f(x - ct) + g(x + ct)$$

- f : right-moving wave
- g : left-moving wave
- Exact decomposition into traveling waves

Incorporating Initial Data

Given

$$u(x, 0) = \phi(x), \quad u_t(x, 0) = \psi(x),$$

solve for f and g :

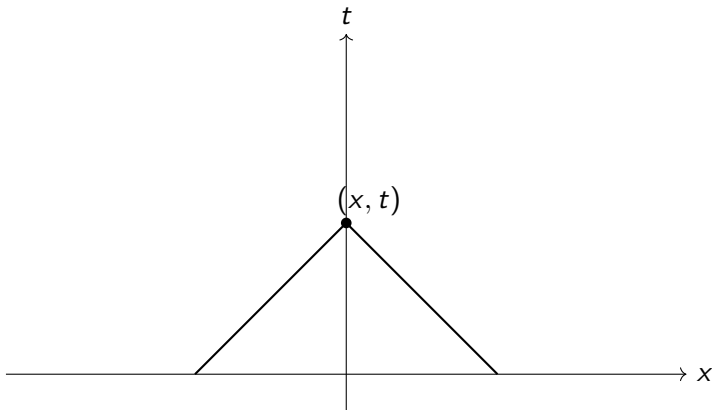
$$f'(x) - g'(x) = \frac{1}{2c}\psi(x),$$

$$f(x) + g(x) = \phi(x).$$

After integration:

$$u(x, t) = \frac{\phi(x - ct) + \phi(x + ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s) ds$$

Geometric Interpretation



Value at (x, t) depends only on data in
 $[x - ct, x + ct]$.

Final Summary

- Burgers: nonlinear geometry $\Rightarrow$ shocks
- Wave: linear geometry $\Rightarrow$ superposition
- Characteristic coordinates reduce wave equation to $u_{\xi\eta} = 0$
- d'Alembert formula is exact traveling-wave decomposition